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ADC-Aware Quantization of LLaMA-Family Models for In-Memory Computing

by

Alina Burykina

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Constructor University | Constructor Institute of Technology

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Abstract

This thesis investigates whether post-training quantization methods can be adapted to deploy LLaMA-family models in in-memory computing systems constrained by low-precision analog-to-digital converters (ADCs). Although prior work such as SmoothQuant and FlatQuant reduces activation and weight quantization error, ADC-constrained inference introduces an additional output quantization bottleneck at the matrix-vector multiplication stage. The work develops an ADC-aware evaluation pipeline for transformer linear layers, studies the limitations of standard post-training quantization in this setting, and explores modifications of FlatQuant that explicitly account for ADC behavior. The experimental part compares naive ADC quantization, baseline FlatQuant, and ADC-aware variants across different tile sizes, quantization precisions, and transform configurations. The main goal is to identify which parts of the LLaMA architecture are most sensitive to ADC constraints, whether post-training quantization alone is sufficient, and under which conditions lightweight adaptation may be required. The outcome of the thesis is intended to clarify the limits of reconstruction-only quantization methods for analog inference and to provide practical guidance for deploying large language models in IMC environments.

Contents

1 Introduction

Purpose of this chapter. Introduce the broader problem of efficient inference for large language models and motivate in-memory computing as a response to the growing cost of both computation and data movement. Explain why ADC precision becomes a bottleneck in IMC systems and why this makes deployment of LLaMA-family models particularly challenging.

What to include. This chapter should explain the practical importance of the topic, define the overall thesis objective, and state why the problem is interesting from both a machine learning and hardware perspective. It should also briefly summarize the gap between conventional digital quantization methods and the specific requirements of ADC-constrained inference.

Suggested structure.

- Motivation: efficient deployment of LLMs.
- Why IMC is attractive for large matrix-vector multiplications.
- Why limited ADC precision creates a distinct quantization problem.
- Why LLaMA-family models are a difficult case.
- A short summary of the thesis contributions.
- A roadmap of the remaining chapters.

Possible contributions paragraph. A concise contribution list at the end of the introduction could state that the thesis: (i) formulates the ADC-constrained quantization problem for LLaMA-family models, (ii) implements an ADC-aware FlatQuant pipeline, (iii) evaluates post-training quantization baselines and ablations, and (iv) analyzes when reconstruction-based PTQ is sufficient and when additional adaptation is needed.

2 Background and Related Work

2.1 In-Memory Computing and ADC-Constrained Inference

Purpose. Provide the hardware context required to understand the rest of the thesis.

What to include. Describe how in-memory computing performs computation directly within memory arrays, why the analog nature of the computation requires an analog-to-digital conversion step, and why ADC precision affects both efficiency and accuracy. This is the right place to define the intuition behind the matrix-vector multiplication pipeline in IMC and to motivate why output quantization is different from standard digital quantization.

2.2 Quantization of Neural Networks

Purpose. Establish the machine learning background.

What to include. Review the main concepts of quantization: post-training quantization (PTQ), quantization-aware training (QAT), symmetric and asymmetric quantization, per-tensor and per-channel quantization, and the straight-through estimator. Keep this section focused on concepts that are directly needed later.

2.3 Challenges of Quantizing Transformer Models

Purpose. Explain why LLMs, and especially LLaMA-family models, are harder to quantize than smaller vision or language models.

What to include. Discuss activation outliers, sensitivity of the residual stream, the role of attention and MLP projections, and why preserving hidden-state geometry is important. You can also explain why some projections, such as `o_proj` and `down_proj`, are often more sensitive to quantization error.

2.4 Existing Methods

Purpose. Position the thesis relative to the state of the art.

What to include. Review methods such as SmoothQuant, FlatQuant, and ADC-oriented ideas such as activation shifting, weight reshaping, bit augmentation, and lightweight adaptation methods. For each method, answer three questions: what problem it solves, why it may help under ADC constraints, and what limitation remains for the current setting.

2.5 Research Gap

Purpose. End the chapter by identifying the precise gap that motivates your own work.

What to include. The main message here can be that existing PTQ methods primarily reduce activation and weight quantization error, whereas ADC-constrained inference introduces a separate output quantization bottleneck that is not explicitly optimized by standard reconstruction-based approaches.

3 Problem Statement and Research Questions

3.1 Problem Definition

Purpose. Formally define the problem studied in the thesis.

What to include. Introduce the linear operation in quantized code space, define the ADC-constrained output quantization step, and explain the roles of the main parameters such as activation precision, weight precision, ADC precision, tile size, and the hardware parameter k. Clearly state that the goal is to preserve model quality under these constraints.

3.2 Research Questions

Suggested research questions.

1. How strongly does limited ADC precision degrade the quality of LLaMA-family models compared with conventional quantization?
2. To what extent can a FlatQuant-style post-training approach mitigate this degradation?
3. Does reconstruction-only PTQ remain sufficient under ADC constraints, or are ADC-aware objectives required?
4. Which layers and projections are most sensitive to ADC quantization?
5. Under what conditions does the problem require lightweight model adaptation beyond PTQ?

3.3 Scope and Assumptions

Purpose. Clarify what the thesis covers and what it does not cover.

What to include. State which LLaMA-family models are evaluated, what hardware behavior is modeled, which parts of the inference pipeline are abstracted, and whether the study focuses on PTQ only or also includes limited adaptation baselines. This subsection is also the right place to state that the work relies on simulation rather than fabricated hardware.

3.4 Evaluation Criteria

Purpose. Define how success will be measured.

What to include. Include both model-level and ADC-level metrics. Model-level metrics may include perplexity or task accuracy. Internal metrics may include layerwise reconstruction error, clipping rate, dead-zone rate, bin utilization, or sensitivity by projection type.

4 Proposed Method

4.1 Post-Training Quantisation

4.1.1 Baseline: FlatQuant for LLaMA

FlatQuant [1] exploits the linear invariance of the matrix-vector product to redistribute the dynamic range of activations before quantisation. For any invertible transform $T \in \mathbb{R}^{d\times d}$,

$$y = xW^\top = \underbrace{(xT^{-1})}_{\tilde{x}}\,\underbrace{(TW^\top)}_{\tilde{W}^\top}.$$

Choosing T to flatten the distribution of $\tilde{x}$ improves both integer quantisation accuracy and ADC bin coverage. Computing a full $d \times d$ transform is prohibitive, so T is Kronecker-factorised as $T = T_1 \otimes T_2$ with $T_1, T_2 \in \mathbb{R}^{\sqrt{d}\times\sqrt{d}}$, reducing the parameter count from $\mathcal{O}(d^2)$ to $\mathcal{O}(d)$.

Calibration proceeds layer by layer in a teacher-student fashion. For each linear projection, a frozen floating-point (FP) teacher produces reference outputs $y^\star = L_{\mathrm{FP}}(x^\star)$, and the quantised student $\hat{y} = \hat{L}_T(\hat{x})$ minimises the mean-squared error $\|\hat{y} - y^\star\|^2$ over the Kronecker factors only, with weights held at their quantised values. After calibration, T is folded into the weights via reparameterisation; inference cost equals that of standard integer quantisation.

We adapt FlatQuant to the LLaMA-3 architecture [2] by wrapping all seven linear projections in each decoder block: q, k, v, o in the grouped-query attention sub-layer, and `gate`, `up`, `down` in the SwiGLU feed-forward network.

4.1.2 ADC Hardware Model and Forward Simulation

In an analog in-memory compute (AIMC) accelerator, quantised weights $\hat{w} \in [-q_w, q_w]^M$ are stored on a crossbar tile of size M. At inference, a quantised activation tile $\hat{x} \in [-q_x, q_x]^M$ is presented, the crossbar accumulates their integer dot product $y_{\mathrm{int}} = \langle \hat{w}, \hat{x} \rangle$ in the analog domain, and an ADC samples the result with a fixed step size δ:

$$\hat{y} = \mathrm{clip}\Big(\Big\lfloor \frac{y_{\mathrm{int}}}{\delta} \Big\rfloor ,\ -2^{b_a - 1},\ 2^{b_a - 1} - 1 \Big) .$$

The step size is a hardware constant set at manufacturing time:

$$\delta = \frac{2\, M\, q_x\, q_w}{2^{b_a}\, k},$$

where b_a is the ADC bit-width and k is the number of ADC channels operated in parallel. Two failure modes follow directly. The *dead zone*: if $|y_{\mathrm{int}}| < \delta$, the floor operation maps the output to zero, discarding the signal entirely. *Clipping*: if $|y_{\mathrm{int}}| > \delta \cdot (2^{b_a - 1} - 1)$, the output saturates. For our target configuration ($b_w = b_x = 4$, $M = 256$, $b_a = 8$, $k = 16$) this gives $\delta \approx 6.12$, coarse enough that a naïve INT4 baseline shows a bypass-to-ADC perplexity gap of $\times$153 (15.41 vs. 2354).

4.1.3 Limitations of Per-Layer Reconstruction

Standard FlatQuant minimises per-projection MSE feeding each layer its *clean* FP activations from the calibration set. Two gaps make this insufficient under ADC inference. First, the calibration objective is blind to the ADC step: a projection optimised in isolation may produce pre-ADC signals that fall inside the dead zone after re-quantisation of the next tile. Second, error accumulates across layers at inference but not during calibration — the input to layer ℓ at inference is the noisy, ADC-distorted output of layer $\ell - 1$, not the clean FP tensor the optimiser saw. These gaps motivate all three ADC-aware extensions described below.

4.1.4 Block-wise Calibration with Propagation and α-Mixing

Block-level reconstruction. Rather than minimising per-projection MSE independently, we reconstruct the output of each entire transformer block:

$$\mathcal{L} = \big\| \hat{B}(\hat{x}) - B_{\mathrm{FP}}(x^\star) \big\|_2^2 .$$

Gradients propagate through the full block ($q/k/v \to \mathrm{attn} \to o \to \mathrm{gate/up} \to \mathrm{down}$), so the transforms of earlier projections are aware of how their error propagates to later ones within the same block.

Propagation across blocks. Blocks are calibrated sequentially from layer 0 to layer $N-1$. When training layer ℓ, its input is the ADC-quantised output produced by the already-trained layer $\ell-1$, not a clean FP activation. Error thus accumulates during calibration in the same way it does at inference, closing the distribution mismatch between training and deployment that per-layer calibration leaves open.

α-mixed objective. A purely ADC-propagated loss produces noisy gradients because the straight-through estimator must pass through multiple floor operations. We interpolate between the FP and ADC losses with a scalar $\alpha \in [0, 1]$:

$$\mathcal{L}(\alpha) = (1 - \alpha)\,\mathcal{L}_{\mathrm{FP}} + \alpha\,\mathcal{L}_{\mathrm{ADC}},$$

where $\mathcal{L}_{\mathrm{FP}}$ uses the clean FP input and $\mathcal{L}_{\mathrm{ADC}}$ uses the quantised input from the previous block. $\alpha = 0$ recovers clean calibration; $\alpha = 1$ is full propagation. An α-sweep (Section 7) shows that $\alpha = 0.5$ minimises ADC perplexity: below 0.5 the loss under-weights the ADC regime; above 0.5 gradient noise degrades transform quality. At $\alpha = 0.5$, INT8 ADC perplexity drops from 28.86 to 14.41 with no additional parameters.

4.1.5 Trainable Per-Channel Diagonals

We first establish a formal bound showing that orthogonal Kronecker transforms cannot suppress per-channel outliers beyond a fixed threshold, and that a diagonal factor is the precise missing degree of freedom.

Lemma 4.1 (Rotation-only outlier suppression bound)**.** *Let $P = L \otimes R$ with $L^\top L = I$, $R^\top R = I$ be an orthogonal Kronecker transform on $\mathbb{R}^d$ ($d = a \cdot b$, $L \in \mathbb{R}^{a\times a}$, $R \in \mathbb{R}^{b\times b}$). For any $x \in \mathbb{R}^d$ define the effective participation ratio*

$$m_{\mathrm{eff}}(x) \;=\; \frac{\|x\|_2^2}{\|x\|_\infty^2} \;\in\; [1,\, d].$$

Then for every orthogonal Kronecker transform P,

$$\|xP\|_\infty \;\geq\; \frac{\|x\|_2}{\sqrt{d}},$$

and consequently the maximum achievable suppression of the peak activation over all such P satisfies

$$\frac{\|x\|_\infty}{\min_P \|xP\|_\infty} \;\leq\; \sqrt{\frac{d}{m_{\mathrm{eff}}(x)}}.$$

Proof. Since P is orthogonal, $\|xP\|_2 = \|x\|_2$. For any vector $y \in \mathbb{R}^d$ the standard ℓ^∞–ℓ^2 inequality gives $\|y\|_\infty \geq \|y\|_2/\sqrt{d}$. Applying this to $y = xP$ yields the first inequality. The suppression ratio bound then follows by dividing $\|x\|_\infty = \|x\|_2/\sqrt{m_{\mathrm{eff}}(x)}$ by the lower bound $\|xP\|_\infty \geq \|x\|_2/\sqrt{d}$. □

Corollary 4.2 (Kronecker capacity)**.** *A full orthogonal matrix on $\mathbb{R}^d$ has $d(d-1)/2$ free parameters. The Kronecker family $L \otimes R$ has only $a(a-1)/2 + b(b-1)/2$ free parameters — for $d = 2048 = 32 \times 64$ this is 2512 vs. $\approx 2.1 \times 10^6$. Consequently, the Kronecker family is a strict low-dimensional subset of all orthogonal transforms and may not attain the bound in Lemma 4.1; the true achievable suppression ratio can be strictly smaller.*

Implications for INT4 ADC. In our setting, per-token activation scaling sets $s_x = \|x\|_\infty / q_{\max}$. If a small number of channels persistently dominate (i.e. $m_{\text{eff}} \ll d$), Lemma 4.1 gives a ceiling on how much rotation alone can help. For example, with $d = 2048$ and 8 dominant channels ($m_{\text{eff}} \approx 8$), the maximum suppression factor is $\sqrt{2048/8} = 16$; if the required suppression exceeds this, rotation-only calibration is provably insufficient. The INT4 baseline confirms this empirically: the ADC perplexity gap of $\times 153$ persists even after FlatQuant rotation, whereas adding diagonal scaling reduces the dead-zone rate from 80.9% to 10.4%.

Motivation and formulation. A Kronecker rotation reshapes the *covariance* of an activation cloud but does not adjust per-channel scales. Under INT4, residual outlier channels remain after rotation: a few channels with magnitudes 5–$10\times$ the median force the per-token scale factor to waste range on those channels and under-represent the rest, which directly worsens the ADC dead-zone rate because y_{int} per output channel depends on how uniformly codes are distributed.

We absorb a trainable diagonal $D = \text{diag}(d_1, \ldots, d_c)$, $d_i \in [10^{-4}, 10]$, into the rotation:

$$\tilde{T} = T \cdot D.$$

The linear map is preserved by applying equal-and-opposite rescaling to the weights:

$$y = \underbrace{(x\, D^{-1} T^{-1})}_{\tilde{x}} \underbrace{(T\, D\, W^\top)}_{\tilde{W}^\top}.$$

The bounds on d_i prevent degenerate scaling. D is trained jointly with T during block-level calibration and folded into $\tilde{W}$ at deployment, adding no inference overhead. Crucially, because D is not orthogonal, it breaks the ℓ^2-energy invariance assumed in Lemma 4.1: setting $d_j < 1$ directly reduces $\|xD\|_2$ in outlier channel j *before* the scale is computed, achieving suppression factors that no rotation can match.

Placement in attention and MLP blocks. We place separate diagonal matrices before the attention sub-layer (`diag_attn`, shared by $q/k/v$, plus a second instance before o) and before the FFN sub-layer (`diag_mlp`, shared by `gate`/`up`, plus a second before `down`). An ablation (Section 7) shows that both placements are necessary: `diag_mlp` drives the bypass-perplexity improvement while `diag_attn` closes the ADC gap; either alone leaves significant perplexity on the table.

4.1.6 Staged Training

Training all Kronecker factors and diagonals jointly from scratch with $\alpha = 0.5$ is unstable: attention gradients dominate early optimisation before MLP transforms have converged, yielding a poor initialisation for both sub-systems.

We split calibration into two stages. *Stage A* (30 epochs, learning rate η) trains only the MLP Kronecker factors $T_{\text{gate}}, T_{\text{up}}, T_{\text{down}}$ and MLP diagonals $D^{(\text{m})}$ with $\alpha = 0$ (clean FP inputs), providing a stable, noise-free landscape in which FFN transforms converge. *Stage B* (10 epochs, learning rate 0.1η) unfreezes the attention Kronecker factors T_q, T_k, T_v, T_o and attention diagonals $D^{(\text{a})}$, and switches to $\alpha = 0.5$. The MLP warm start prevents gradient interference from attention. This procedure achieves the best INT4 results among all pure-PTQ configurations: bypass perplexity 18.50, ADC perplexity 27.60.

4.2 Quantisation-Aware Adaptation via Post-ADC Residual LoRA

PTQ methods reach a floor around 27.5–28.5 ADC perplexity for INT4 regardless of calibration strategy. The limiting factor is ADC resolution: with $\delta \approx 6.12$ each output channel has at most $\sim$30 usable discrete levels, introducing irreducible rounding error that no linear transform can eliminate. We complement PTQ with a lightweight adaptation step that learns to correct ADC distortion *after* it has occurred, in the digital domain, without modifying the frozen quantised model.

4.2.1 Architecture

The PTQ checkpoint is frozen entirely. For each of the seven linear projections in every decoder block, we add a parallel low-rank branch that reads the full-precision input x_{fp} and adds its output to the ADC result:

$$y = \underbrace{\text{ADC}_{\text{frozen}}(\hat{x})}_{\text{quantised path}} + \underbrace{\frac{\alpha_r}{r} B(A(x_{\text{fp}}))}_{\text{residual correction, rank } r} ,$$

where $A \in \mathbb{R}^{r \times d}$ and $B \in \mathbb{R}^{d \times r}$ are LoRA [3] factors stored and computed in `float32`, and α_r / r is the standard LoRA scaling. All base model parameters — weights, Kronecker factors, diagonals, and ADC step sizes — remain frozen throughout. For Llama-3.2-1B ($d = 2048$, 16 decoder blocks, 7 projections each), rank-4 adapters add $\sim$1.8M parameters (0.15% of the total model).

4.2.2 Training Objective

The adapters are trained for 5 epochs with learning rate 10^{-4} on the same 128-sample calibration set used for PTQ. The loss combines cross-entropy on the model's predictions with a knowledge-distillation term towards the FP teacher:

$$\mathcal{L}_{\text{LoRA}} = \text{CE}(\hat{p},\, y) + \lambda\, \text{KL}\big(\hat{p} \,\big\|\, p_{\text{FP}}\big) ,$$

with $\lambda = 1$. The FP teacher forward pass is computed once per batch with no gradient. Ablations (Section 7) show that the KL term improves ADC perplexity by $\sim$0.6 points over CE alone, and that rank $r = 4$ saturates the gain — rank 8 yields only marginal further improvement. Restricting adapters to the last eight layers is significantly worse than covering all sixteen, confirming that ADC distortion accumulates across the full depth of the network.

4.2.3 Why Post-ADC Placement

Placing the LoRA correction *before* the ADC — modifying the pre-ADC integer signal — fails catastrophically. Any correction $\Delta = BAx$ is itself quantised by the fixed step δ before reaching the output, so it cannot express sub-δ adjustments. Moreover, gradients through the floor operator are too noisy for stable training from a cold start, and the coupled interaction with the hard ADC clamp causes divergence.

Post-ADC placement avoids both problems: the correction is added in the digital domain after the ADC read, at full floating-point precision, with clean gradients from the CE+KL loss. ADC perplexity drops from the PTQ plateau of 27.60 to 13.96 for INT4 on WikiText-2, surpassing the INT8 PTQ result (14.41), and generalises to C4 (ADC perplexity 23.30 vs. PTQ baseline 46.40).

5 Implementation

5.1 Software Architecture

Purpose. Describe the codebase at a system level.

What to include. Explain the structure of the implementation, including transformation modules, quantization modules, calibration logic, and model-level utilities. A simple pipeline figure would work well here.

5.2 Core Components

Suggested components to describe.

- `KroneckerTransform` and its role in reparameterization.
- `FlatQuantLinear` as the main quantized linear wrapper.
- The ADC-aware simulation path for tiled matrix-vector multiplication.
- The LLaMA attention and MLP wrappers.

5.3 Calibration Pipeline

Purpose. Explain how calibration is executed in practice.

What to include. Describe how input activations are captured, how decoder layers are processed sequentially, how floating-point reference outputs are computed, which parameters are trainable during calibration, and how outputs are propagated from one layer to the next.

5.4 Numerical Stability and Engineering Decisions

Purpose. Document important implementation choices.

What to include. This subsection can cover mixed precision, float32 accumulation, gradient clipping, clamping of transform parameters, NaN checks, and memory-management decisions. These practical details are valuable in a thesis because they justify why the implementation is reliable.

5.5 Reproducibility

Purpose. Make the work easy to reproduce.

What to include. State the software versions, hardware environment, random seed policy, configuration management, and how experiment outputs are logged and stored.

6 Experimental Design

6.1 Model, Data, and Hardware Setup

Model. All experiments use Llama-3.2-1B [2] in `bfloat16`. The model has 16 decoder layers, hidden dimension $d = 2048$, 32 attention heads with grouped-query attention, and a SwiGLU FFN with intermediate dimension 8192. It is small enough to run a full sweep of dozens of configurations overnight on a single GPU while being representative of the transformer architecture used in larger LLaMA-family models.

Calibration data. FlatQuant transforms and LoRA adapters are calibrated on WikiText-2 training text [4]. Unless stated otherwise, 128 randomly sampled sequences of 2048 tokens each are used. Selected experiments use 512 or 1024 samples to study the effect of calibration set size; the INT4 series demonstrates that sample count is a dominant lever for this bit-width.

Evaluation. Perplexity is measured on the WikiText-2 test set using a sliding window of context length 2048 and stride 1024. Selected configurations are additionally evaluated on C4 [5] under the same protocol to verify out-of-domain generalisation.

ADC hardware parameters. Table 1 lists the fixed hardware constants used throughout. All values are chosen to match a plausible near-term AIMC chip target. The ADC step size δ is fully determined by these constants via $\delta = 2Mq_xq_w/(2^{b_a}k)$, giving $\delta \approx 2016$ for INT8 and $\delta \approx 6.12$ for INT4.

Table 1: Fixed ADC hardware parameters.

Parameter	Symbol	Value
Tile size (crossbar columns)	M	256
ADC bit-width	b_a	8
Parallel ADC channels	k	16
INT8 weight range	q_w	127
INT8 activation range	q_x	127
INT4 weight range	q_w	7
INT4 activation range	q_x	7
ADC step (INT8)	δ	≈ 2016
ADC step (INT4)	δ	≈ 6.12

FlatQuant calibration settings. Transforms are trained for 30 epochs with learning rate 5×10^{-3} using the AdamW optimiser. Stage B of staged training uses 10 epochs at 5×10^{-4}. Static activation scales for Step 2 of the FlatQuant pipeline use the 99.9th-percentile clipping strategy (`calibration_method=percentile`).

6.2 Evaluation Metrics

All experiments are characterised by the following metrics.

- **PPL (bypass).** WikiText-2 perplexity with ADC disabled; only tiling and integer quantisation are active. Measures FlatQuant transform quality in isolation. Lower is better; the FP baseline is ≈ 10.5 (INT8 regime) and ≈ 10.5 / ≈ 14 depending on bit-width.
- **PPL (ADC).** WikiText-2 perplexity with the full ADC pipeline enabled. This is the primary metric; the bypass–ADC gap measures the additional degradation introduced by finite-resolution ADC.
- **dead_rate.** Fraction of output channels across all layers and calibration tokens for which $|y_{\text{int}}| < \delta$, i.e. the ADC output is identically zero. Measured on the calibration set; lower is better.
- **clip_rate.** Fraction of ADC outputs that saturate at the maximum code $\pm(2^{b_a-1}-1)$. High clip rate indicates the integer accumulation systematically overflows the ADC range.
- **bin_usage.** Fraction of the 2^{b_a} ADC output bins that are actually occupied, averaged over layers and tokens. Higher is better; near-zero bin usage indicates the ADC is effectively acting as a 1-bit comparator.
- **reconstruction_rel.** Relative mean-squared error between the ADC output and the FP reference, $\|\hat{y}-y^\star\|^2/\|y^\star\|^2$, on the calibration set. Tracks layer-level signal fidelity independently of perplexity.

6.3 Baseline Configurations

FP baseline. The original `bfloat16` Llama-3.2-1B with no quantisation, giving WikiText-2 perplexity ≈ 10.5. This is the upper bound on achievable quality.

FlatQuant w8a8 PTQ baseline (INT8). FlatQuant with $b_w = b_x = 8$, 128 calibration samples, per-token activation scaling (`calibration_method=percentile`), and ADC enabled at inference. This configuration gives bypass PPL 10.01 and ADC PPL 28.86 with dead_rate 10.3%. It is the reference point for all INT8 ablations.

FlatQuant w4a4 PTQ baseline (INT4). FlatQuant with $b_w = b_x = 4$, 128 calibration samples, same ADC hardware parameters. After correcting a bug in the LLaMA ADC converter that was creating spurious uncalibrated copies of attention projections, this gives bypass PPL 15.41 and ADC PPL 2354.86, a bypass-to-ADC gap of $\times 153$. This extreme gap motivates the INT4-specific extensions.

6.4 PTQ Experiment Series

Experiments are grouped into versioned overnight sweeps. Each sweep trains from scratch and evaluates all configurations on the same calibration and evaluation sets; results are logged to JSON files and WandB (project `llama-flat-quant-adc-ptq-blocks`).

INT8 calibration study (v2). Three INT8 configurations are compared: (i) *baseline* — per-projection MSE, $\alpha = 0$; (ii) *bin-center* — adds a $\cos^2(\pi z)$ auxiliary loss ($z = y_{\text{int}}/\delta$) with $\lambda = 0.1$ to nudge accumulations toward bin centres, reducing floor-rounding error;

(iii) *propagated* — block-wise calibration, $\alpha = 1$; (iv) *prop+center* — combines propagation with the bin-center auxiliary loss. The goal is to isolate the contribution of each component to the bypass–ADC gap.

INT4 propagation and α-sweep (v1 and v2). Version 1 sweeps eight configurations at 128 and 512 calibration samples to identify the effect of sample count and propagation for INT4. Version 2 introduces the α-mixing objective and sweeps $\alpha \in \{0.25, 0.5, 0.75, 1.0\}$ at 1024 samples. It also evaluates the two-stage protocol (§4.1.6) and the bounded-diagonal extension (`add_diag`, $d_i \in [10^{-4}, 10]$) combined with $\alpha = 0.5$.

Layer-wise α and selective diagonals (v3). Two additional degrees of freedom are introduced: (i) *layer-wise* α uses a smaller α_{early} for the first $N/2$ layers and a larger α_{late} for the rest, on the hypothesis that early layers accumulate more ADC error; (ii) *selective diagonals* trains `diag_attn` only, `diag_mlp` only, or both, to isolate which sub-layer benefits most from per-channel rescaling. Configurations are crossed with flat and layer-wise α.

MLP diagonal decomposition and staged selective training (v4). The MLP diagonal is split into `diag_up_gate` and `diag_down` components to determine which projection drives the bypass improvement seen in v3. A two-stage protocol is also evaluated: Stage A trains MLP diagonals without propagation ($\alpha = 0$); Stage B adds attention diagonals and switches to $\alpha = 0.5$.

Stochastic propagation (v5). Deterministic $\alpha = 0.5$ mixes FP and ADC losses in a fixed ratio every batch. Two stochastic alternatives are evaluated: (i) *Bernoulli* — each batch randomly uses either a clean FP input or an ADC-propagated input with probability 0.5; (ii) *Beta(2,2)* — each batch samples $\alpha \sim \text{Beta}(2, 2)$ and performs a dual forward pass with the sampled mix. Both modes are tested with the staged protocol from v4.

6.5 QAT Experiment Series: Post-ADC Residual LoRA

Initial LoRA sweep (v6). Starting from the best PTQ checkpoint (`staged_mlpdiag_then_attn`, ADC PPL 27.60), LoRA adapters are added with $r \in \{4, 8\}$ and trained on WikiText-2 calibration data for 5 epochs with cross-entropy loss. Four target sets are compared: `down_proj` only, `down + o_proj`, all 7 projections, and no LoRA (control).

Ablation study (v7). Six ablation axes are evaluated from scratch:

- *Seed reproducibility* (group A): three independent seeds for the full PTQ+LoRA pipeline to quantify end-to-end variance.
- *Rank sweep* (group B): $r \in \{1, 2, 4, 8\}$ with `down + o_proj` targets to locate the saturation point.
- *Loss function* (group C): cross-entropy (CE) vs. CE + KL divergence from the FP teacher ($\lambda = 1$).
- *Layer coverage* (group D): all 16 layers vs. first 8 vs. last 8, with rank 4 and `down + o_proj`.

- *Placement* (group E): post-ADC residual (proposed) vs. pre-ADC injection (modifying the integer accumulation before the ADC floor operation).
- *Full coverage with CE+KL* (group F): all 7 projections with both loss variants.

Generalisation evaluation (v8). The top-3 LoRA configurations from v7 are re-trained and evaluated on both WikiText-2 and C4 to test whether the correction learned on in-domain calibration data transfers to out-of-domain web text.

Tile size comparison. A dedicated sweep compares $M \in \{256, 1024\}$ without and with the best LoRA configuration (`r4_all_ce_kl`). Changing M scales δ by the same factor, coarsening ADC resolution for larger tiles; this tests whether LoRA correction remains effective under a different hardware regime.

7 Results

7.1 Baseline Results

Purpose. Present the main quantitative comparison.

What to include. Compare floating-point inference, naive ADC conversion, and baseline FlatQuant. This is where the reader first sees whether the proposed direction is promising at all.

7.2 Effect of Hardware Parameters

Purpose. Analyze how model quality changes with the ADC setup.

What to include. Show the effect of tile size, ADC bit precision, and any other hardware parameters you vary. Graphs are especially useful here because trends matter more than isolated numbers.

7.3 Effect of Method Components

Purpose. Show which parts of the method are responsible for improvements.

What to include. Present the ablation results for transform type, transform scope, clipping options, and ADC-aware losses. This subsection is essential for arguing that the method is not just a bundle of arbitrary design choices.

7.4 Layer and Projection Sensitivity

Purpose. Provide a more detailed analysis of where ADC error hurts most.

What to include. This is the right place to compare attention and MLP sensitivity and to discuss whether certain projections such as `o_proj` and `down_proj` dominate the quality degradation.

7.5 Best Configurations

Purpose. Summarize the strongest final results.

What to include. Present the best-performing PTQ configuration, and if applicable, the best lightweight adaptation result. A final summary table is helpful here.

8 Discussion

8.1 Interpretation of Results

Purpose. Move from raw results to explanation.

What to include. Discuss why certain modifications work and others do not. Relate the observed behavior back to the theory of ADC-constrained output quantization and the internal statistics of LLaMA layers.

8.2 Limits of PTQ Under ADC Constraints

Purpose. State clearly where the chosen approach succeeds and where it fails.

What to include. This subsection should answer whether reconstruction-based PTQ remains sufficient or whether the problem fundamentally requires some degree of model adaptation in the low-bit regime.

8.3 Practical Implications

Purpose. Explain what the results mean beyond the specific experiments.

What to include. Discuss what your findings imply for IMC deployment, for the design of ADC-aware quantization methods, and for future hardware-software co-design of transformer inference systems.

8.4 Threats to Validity

Purpose. Strengthen the scientific quality of the thesis.

What to include. Mention limitations such as reliance on simulation, restricted model sizes, calibration dataset choice, and the gap between simplified hardware assumptions and a real chip implementation.

9 Conclusion

Purpose. Provide a concise answer to the research questions and summarize the thesis contributions.

What to include. Restate the original problem, summarize the method and experimental findings, state the main conclusion about ADC-aware PTQ for LLaMA-family models, and list a few concrete directions for future work such as better output-aware objectives, mixed precision, or lightweight adaptation.

References

[1] Yuxuan Chen et al. “FlatQuant: Flatness Matters for LLM Quantization”. In: *arXiv preprint arXiv:2410.09426* (2024). URL: https://arxiv.org/abs/2410.09426.

[2] Abhimanyu Dubey et al. “The Llama 3 Herd of Models”. In: *arXiv preprint arXiv:2407.21783* (2024). URL: https://arxiv.org/abs/2407.21783.

[3] Edward J. Hu et al. “LoRA: Low-Rank Adaptation of Large Language Models”. In: *International Conference on Learning Representations (ICLR)*. 2022. URL: https://arxiv.org/abs/2106.09685.

[4] Stephen Merity et al. “Pointer Sentinel Mixture Models”. In: *International Conference on Learning Representations (ICLR)*. 2017. URL: https://arxiv.org/abs/1609.07843.

[5] Colin Raffel et al. “Exploring the Limits of Transfer Learning with a Unified Text-to-Text Transformer”. In: *Journal of Machine Learning Research* 21.140 (2020), pp. 1–67. URL: https://arxiv.org/abs/1910.10683.

A Detailed Experimental Configurations

Suggested contents. Put complete hyperparameter tables here: number of calibration samples, batch size, learning rates, transform settings, clipping settings, hardware parameters, and final selected configurations for each run.

B Additional Figures and Tables

Suggested contents. Add extended ablation tables, per-layer plots, extra sensitivity analyses, and any visualizations that support the main text but are too large to place in the core chapters.

C Code Organization and Reproducibility Notes

Suggested contents. Briefly describe the repository structure, entry points for training and evaluation, external dependencies, and the exact steps required to reproduce the reported results.